POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2017
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) How aliasing is encountered in the processing of discrete-time signals? What can be done to minimize aliasing effect? Explain with the basic elements of digital signal processing where aliasing can be cancelled. 7

b) A digital communication link carries binary-coded words representing samples of an input signal 8

$x_a(t) = 3\cos 5000\pi t + 2\sin 6000\pi t$

The link is operated at 18,000 bits/s and each input sample is quantized into 512 different voltage levels.

i. What is the discrete-time signal obtained after sampling?

ii. What is the resolution Δ?

2. a) Prove that multiplication of a sequence x[n] by $e^{jw_0 n}$ is equivalent to a frequency translation of the spectrum $X(e^{jw})$ by w_0. 7

b) An LTI, causal, second order discrete-time system is described by the difference equation 8

$$y[n] = 6y[n-1] - 8y[n-1] + 16x[n]$$

Calculate its step response s[n].

3. a) Prove that necessary and sufficient condition for stability is $\sum_{-\infty}^{\infty} h[k] < \infty$. 7

b) Why is DFT preferred over DTFT in the analysis of discrete - time signals? Also determine the DFT of the signal x[n] = u[n] - u[n-4] using DIF-FFT algorithm. 8

4. a) What is the difference between linear and circular convolution? Find the linear circular convolution of the following sequences: 7

x[n] = {1, 0, 0 ,1} and h[n] = {2,0,2}

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Or

Show that multiplication of two discrete time signals results in circular convolution of their DFTs

b) Draw the lattice-ladder structure for the following IIR system. 5

$$H(z) = \frac{1 - 0.8z^{-1} - 0.15z^{-2}}{1 + 0.1z^{-1} - 0.72z^{-2}}$$

Is the system stable?

5. a) Design a digital Butterworth low pass filter satisfying the constraints 7

$$\begin{cases} -3.011dB \le |H(e^{jw})| \le 0dB & 0 \le w \le \frac{\pi}{2} \\ |H(e^{jw})| \le 10.45dB & \frac{3\pi}{4} \le w \le \pi \end{cases}$$

With T = 1sec using bilinear transformation method. Realize the filter using the most convenient realization form.

b) Given an analog system 7

$$H_a(s) = \frac{s^2 + 1.4s + 9.6}{(s + 0.5)(s^2 + 0.4s + 9.4)}$$

Obtain a digital filter by using impulse invariant method. Assume T = 1.

6. a) Explain briefly the design of linear phase FIR filter by frequency Sampling method with proper example 6.46

b) Design a linear FIR filter using Kaiser window to meet the following specifications: 6.3

$$0.98 \le |H(e^{j\omega})| \le 1.02, \quad for\ 0 \le |\omega| \le 0.2\pi$$
$$|H(e^{j\omega})| \le 0.03, \quad for\ 0.3\pi \le |\omega| \le \pi$$

7. Write short notes on: **(Any two)**
   a) Properties of LTI system. 2
   b) Computational complexity of DFT. 4
   c) Limit Cycle Oscillation Effect. ?

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